

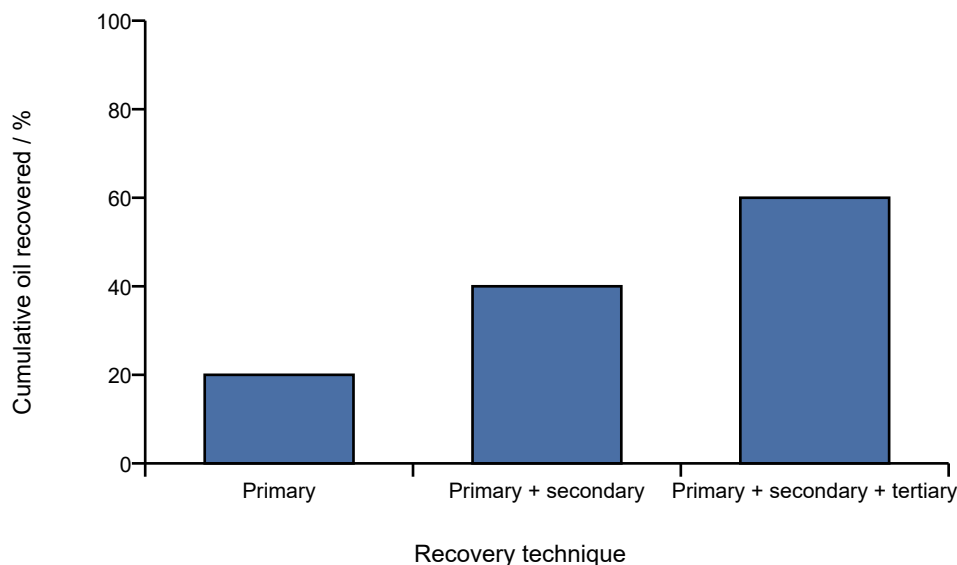
## Questions by topic: Fossil fuels & nuclear power: sustainability and new technologies - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1

Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 1

1

Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 1

2

Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 1

3

Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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**0 1 . 4** Explain **three** advantages of directional drilling. **[3 marks]**

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**10**

**0 2 . 1** Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction. **[3 marks]**

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**0 2 . 2** Describe **four** environmental concerns about fracking and suggest how they may be reduced. **[5 marks]**

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Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

**[2 marks]**

10

**03.1**

Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

**[4 marks]**

**0 3 . 2** Describe what methane hydrate is and where it is found.

**[2 marks]**

**0 3 . 3** Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

**[4 marks]**

**10**

**0 4** Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

**Table 1**

| Feature                               | Coal-fired         | Nuclear (uranium)                            |
|---------------------------------------|--------------------|----------------------------------------------|
| Fuel used per year                    | 3.5 million tonnes | 30 tonnes                                    |
| CO <sub>2</sub> emitted per kWh / g   | 900                | 12 (mostly fuel processing and construction) |
| Construction cost / £ billion         | 2                  | 20                                           |
| Fuel cost per kWh / pence             | 3.0                | 0.5                                          |
| Typical response to changes in demand | Hours              | Slow (days)                                  |

**0 4 . 1** Explain how nuclear fission releases energy in a reactor.

**[3 marks]**

0 4 . 2

Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

0 4 . 3

Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

0 4 . 4

Explain why the CO<sub>2</sub> emissions of nuclear power are not zero.

[1 mark]

0 4 . 5

Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

1 .....

2 .....

10

05 . 1

Describe **three** factors that have restricted the growth of nuclear power.  
[3 marks]

- 1 .....
- 2 .....
- 3 .....

05 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.  
[3 marks]

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05 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.  
[4 marks]

**Advantages** .....

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**Disadvantages** .....

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06 . 1 Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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06 . 2 State the sources of the two fuels used in fusion research.

[2 marks]

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2 .....

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06 . 3 Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

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06 . 4 Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

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07.1

Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

Extra space

07 . 2

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

10

**END OF QUESTIONS**